

[Self] Energy consumption of PC gaming vs ChatGPT

Posted by u/TheMisterTango • 0 points • 18 comments

I see a lot of talk about AI energy consumption so I thought a bit of context is warranted. From steamcharts, there are currently approximately 3.6 million concurrent players for the top 10 most popular steam games. There are varying figures for the power draw of consumer PCs, so I'll take something on the lower end of the range of what I've seen and say the average consumer gaming PC draws 250 watts during gameplay. 3.6 million people drawing an average of 250 watts comes out to 900 MW of power consumption, I'll assume it's fairly consistent and call that 900 megawatt-hours concurrent. Extrapolate that over a 24 hour day and you get 21,600 megawatt-hours of power consumption. Compare that to ChatGPT, which uses an estimated 500 MW per day. That means that just the top 10 games on steam use more energy in a single hour than ChatGPT uses in an entire day. For a whole day, the top 10 steam games draw 43x as much energy as ChatGPT.

Comments

u/doublej42 • 3 points • 2025-04-13 14:59 UTC

Running nothing locally other than like checking my email is 250 watts if you count my monitors and mine as pretty power efficient at 70 watts each.

Gaming I get up to 660 watts. Running a local AI I get around 500w.

The big power use if training which is a lot but still pales in comparison to stuff like metal smelting.

AI does eventually have the advantage of nuclear power and no loss from the grid resistance.

u/echoingElephant • 2 points • 2025-04-13 14:55 UTC

Dude, sorry, but you made a huge mistake. Assuming your main math is correct, computers draw 900MW of power. That is not an amount of energy. It's power.

But now you say that 900MW come out to 21600MWh over a day. Sure.

However, you somehow decided to compare that figure, 21600MWh, to the power draw of ChatGPT to get to a factor of 43. Yet, obviously, this is just bs. You compare energy to power. You need to compare 900MW to 500MW for ChatGPT (assuming your data is correct). Your mistake becomes visible in you saying that ChatGPT supposedly uses 500MW „per day“.

The factor therefore is 1.8, NOT 43.

u/TheMisterTango • -1 points • 2025-04-13 15:02 UTC

How is it wrong? 900 megawatts power draw concurrent times 24 hours in a day is 21,600 megawatt-hours, that's not a debate.

u/echoingElephant • 2 points • 2025-04-13 15:13 UTC

And just to point out the obvious: ChatGPT doesn't use 500MWh. According to the Washington Post, it uses 40 million kWh a day, or 40,000MWh.

This is double of what you calculated for the Steam users.

u/echoingElephant • 2 points • 2025-04-13 15:11 UTC

What? No, you can't. Absolutely not.

The computers draw 900MW of power at all times. Every minute. So $900\text{MW} \times 24\text{h}$ gives you the figure in MWh.

ChatGPT, you. Say, uses 500MW every hour. That's the same number as the one for the computers. ChatGPT doesn't use 500MWhs of energy. It uses $500\text{MW} \times 24\text{h}$.

You absolutely cannot compare numbers in two different units and then claim a dimensionless ratio. That's unscientific and utter nonsense.

u/TheMisterTango • -2 points • 2025-04-13 15:18 UTC

It was a typo, I meant to say you cannot. And I didn't say ChatGPT uses 500MW every hour, I said every day, according to [this source](<https://www.forbes.com/sites/cindygordon/2024/03/12/chatgpt-and-generative-ai-innovations-are-creating-sustainability-havoc/#>).

u/Hour_Ad5398 • 2 points • 2025-04-13 15:28 UTC

>And I didn't say ChatGPT uses 500MW every hour,

you cant use "500mw every hour". watt is a momentary measurement. if you use 500mw for a second, its 500mws. if you use it for an hour, its 500mwh.

u/TheMisterTango • 0 points • 2025-04-13 15:29 UTC

I know how units work, which is why I used MWh in my post.

u/Hour_Ad5398 • 1 points • 2025-04-13 16:53 UTC

>And I didn't say ChatGPT uses 500MW every hour, I said every day,

who said this then? me?

u/TheMisterTango
• 0 points •
2025-04-13
16:54 UTC

Idk, you
pulled it out
of your ass
I guess
because I
surely
never said
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u/echoingElephant • 2 points • 2025-04-13 15:23 UTC

Your source is wrong. That shows in them using „kilowatts“ as a unit of energy.

u/echoingElephant • 3 points • 2025-04-13 15:22 UTC

IT CANNOT USE 500MW PER DAY. Megawatts are a measure of power draw. Not energy.

This is as if you said „My car uses 10l of gasoline per 100km per day“. It's nonsensical (in fact it means that it uses more fuel per 100km the more days you have).

The gamers use 20,000 kWh per day. ChatGPT uses 500MW*24h per day.

u/TheMisterTango • -1 points • 2025-04-13 15:26 UTC

I know how units work, blame the guy who wrote the article. It just says 500k kw/day, so I just assumed they actually meant kwh because the guy who wrote it doesn't know the difference.

u/echoingElephant • 2 points • 2025-04-13 15:30 UTC

You copied it incorrectly too. And it appears as if they wanted to write kilowatts, as in the unit of power, and you incorrectly assumed that it was energy because they added „per day“.

u/TheMisterTango
• 0 points •
2025-04-13
15:40 UTC

If you don't like the source I used then here's a different one that's a bit more precise with it's units:
[source]
(<https://balkangreenenergynews.com/consumes-enough-power-in-one-year-to-charge-over-three-million-electric-cars/>). This one says it uses 226.8 GWh per year, which comes out to about 621 MWh per day.

